



5 The curve C has polar equation $r^2 = \tan 2\theta$, where $0 \leq \theta \leq \frac{1}{8}\pi$.

(a) Sketch C and state the greatest distance of a point on C from the pole.

[2]

.....

(b) Find the exact value of the area of the region bounded by C and the half-line $\theta = \frac{1}{8}\pi$.

[4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....





- (c) Show that C has Cartesian equation $x^4 - 2xy - y^4 = 0$ given that $0 \leq x \leq \cos\left(\frac{1}{8}\pi\right)$ and $0 \leq y \leq \sin\left(\frac{1}{8}\pi\right)$.

[4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (d) Using your answer to (b), deduce the exact value of the area bounded by C , the x -axis and the line $x = \cos\left(\frac{1}{8}\pi\right)$.

[2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

